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| **Fiscal Year:** | FY 2014 | **Task Last Updated:** | FY 10/23/2014 |
| **PI Name:** | Delp, Michael Ph.D. | | |
| **Project Title:** | Disuse Osteopenia: A Potential Vascular Coupling Mechanism | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | mdelp@fsu.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 352-214-1195 |
| **Organization Name:** | Florida State University | | |
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| **Zip Code:** | 32306-0001 | **Congressional District:** | 2 |
| **Comments:** | Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005). | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2011 Space Biology NNH11ZTT002N |
| **Start Date:** | 08/01/2012 | **End Date:** | 07/14/2014 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Smith, Jeffrey | **Contact Phone:** | 650-604-0880 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 7/14/2014 (from 7/31/2015) per PI move to Florida State University. Project will continue under grant NNX14AQ57G. Info per A. Chu/ARC and NSSC (Ed., 9/23/14) | | |
| **Key Personnel Changes/Previous PI:** | None | | |
| **COI Name (Institution):** | Allen, Matthew ( Indiana University School of Medicine )<br>Delp, Judy ( University of Florida ) | | |
| **Grant/Contract No.:** | NNX12AL41G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Disuse osteopenia, which is associated with habitation in a microgravity environment, is a condition of low bone density resulting in increased fragility and greater likelihood for bone fracture. Although this condition is largely attributed to the direct effects of unloading of the bone, coupling mechanisms associated with diminished bone and marrow perfusion, attenuated fluid flow within the bone, and fewer signaling molecules from vascular cells may also play a modulating role on the bone loss associated with spaceflight.<br>In the tail-suspended hindlimb-unloaded (HU) rat, an animal model of microgravity-induced osteopenia, hindlimb bone and marrow blood flow decreases with acute unloading (10 min) and continues to progressively decline through 28 days of unloading. The initial and chronic decrements in perfusion are primarily due to increases in osseous vascular resistance. In addition, when the HU rat hindlimb skeleton is reloaded during standing, long bone and marrow perfusion are lower and vascular resistance higher than that in standing control animals. These higher vascular resistances in HU rats could result from, 1) increased vasoconstrictor responses of bone resistance arteries, or 2) decreased endothelium-dependent and/or endothelium-independent vasodilator responsiveness of bone resistance arteries.<br><br>The general hypothesis of this proposal is that skeletal unloading induces alterations in skeletal resistance artery function; these alterations serve to diminish bone perfusion and ultimately disrupt mechanisms for maintaining the normal balance between bone formation and bone resorption. The specific purpose is to elucidate vasoconstrictor and vasodilator mechanisms involved in the control of bone and marrow perfusion, including endothelial signaling pathways forming nitric oxide and prostacyclin, in bone during conditions of unloading, and to determine whether deficits in bone blood flow and endothelium-dependent vascular signaling are related to unloading-induced bone loss. To accomplish these goals, the principal nutrient artery of the femur from control and 14 day HU rats will be isolated and cannulated for in vitro experimentation. Mechanisms of smooth muscle and endothelial cell function in the osseous circulation will be determined, as well as how skeletal unloading alters these vasoconstrictor and vasodilator control mechanisms (Aims 1-4). Studies will then be conducted using adenoviral gene transfer (Aim 5) to manipulate bone and marrow perfusion and endothelium-dependent vasodilation to determine whether blood flow and vascular endothelial cell signaling is coupled to disuse-associated bone remodeling in the femur (i.e., bone formation rates, mass, total, cortical and cancellous bone mineral density, and mechanical strength). These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders which occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.<br><br>EDITOR's NOTE (9/23/2014): Project will continue under grant NNX14AQ57G with the same title due to PI move to Florida State University, from University of Florida. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders which occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age. |
| **Task Progress:** | One manuscript has been published and a second has been submitted for review. In the case of the first published manuscript (Stabley JN, Prisby RD, Behnke BJ, Delp MD. Chronic skeletal unloading of the rat femur: mechanisms and functional consequences of vascular remodeling. Bone. 2013 Dec;57(2):355-60), the purpose of this investigation was to determine, 1) whether vascular remodeling occurs in the principal nutrient artery (PNA) of the rat femur following 7- and 14-d of skeletal unloading, 2) whether alterations in PNA circumferential wall stress or intraluminal shear stress are regulated variables when putative changes in vascular structure occur, and 3) whether possible changes in vascular structure are associated with alterations in bone blood flow capacity with reloading. It was hypothesized that unloading-induced reductions in hindlimb arterial pressure would diminish PNA circumferential wall stress while reductions in femoral blood flow would reduce PNA intraluminal shear stress. Further, reductions in these mechanical stresses would be normalized through decreases in PNA medial wall thickness and intraluminal diameter, respectively. Additionally, we hypothesized that the shear stress-induced reductions in PNA intraluminal diameter would be associated with a diminished bone blood flow capacity with reloading. The results (see Bibliography for publication information) provided the first evidence that chronic skeletal disuse alters bone and marrow blood flow capacity. Specifically, following 7 d of unloading there was diminished blood flow with reloading in a single region of the femur, the distal metaphysis. When the duration of unloading was increased to 14 d perfusion was diminished in all regions of the femoral bone and marrow during reloading. These reductions in blood flow following 14 d unloading were the result of elevations in vascular resistance within the bone and marrow and were associated with a remodeling of the resistance vasculature. This remodeling involved changes in both maximal diameter and wall thickness of the PNA. For example, the diameter of the PNA was ~15% lower with 14 d HU and the PNA medial wall thickness was ~21% less than that in Con rat PNAs. While the remodeling of the bone vasculature served to normalize the intraluminal shear stress and circumferential wall stress associated with the unloading-induced changes in bone blood flow and arterial pressure, respectively, the structural narrowing of the resistance vasculature also could function to elevate vascular resistance and limit perfusion of bone and marrow during reloading.<br>Data contained in the second submitted manuscript (Prisby, RD, BJ Behnke, MR Allen, and MD Delp. Effects of skeletal unloading on the vasomotor properties of the rat femur principal nutrient artery. Submitted) tests functional changes to the femoral PNA vasoconstrictor (Specific Aim #1) and vasodilator (Specific Aims #2 & #4) properties. PNA vasoconstrictor responsiveness to myogenic norepinephrine, clonidine, KCl, endothelin, and arginine vasopressin were unchanged by 14 d unloading, whereas vasoconstriction induced by phenylephrine was diminished. These data indicate that vasoconstrictor responsiveness is not enhanced in the PNA, and therefore, altered intrinsic vasoconstrictor responsiveness does not contribute to the decrease in bone and marrow blood flow and increase in vascular resistance during reloading. In addition, changes in PNA endothelium-independent vasodilation to the NO-donor sodium nitroprusside and the metabolite adenosine were unaffected by HU. However, PNA endothelium-dependent vasodilation, such as that elicited through acetylcholine and flow, were lower in HU rats relative to that in cage controls. This change in endothelium-dependent vasodilation was mediated through the NO signaling pathway and not the COX mechanism. Such changes in endothelium-dependent vasodilation could contribute to the reduction in bone and marrow blood flow and increase in vascular resistance during reloading. Furthermore, the decrements in endothelium-dependent vasodilation are significantly related to changes in cancellous bone mineral density in the femur. |

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| | EDITOR's NOTE (10/20/2014): Project will continue under grant NNX14AQ57G with the same title due to PI move to Florida State University, from University of Florida. See that project for subsequent reporting. |
| **Bibliography Type:** | Description: (Last Updated: 07/09/2025) |
| **Articles in Peer-reviewed Journals** | Stabley JN, Prisby RD, Behnke BJ, Delp MD. "Chronic skeletal unloading of the rat femur: mechanisms and functional consequences of vascular remodeling." Bone. 2013 Dec;57(2):355-60. Epub 2013 Sep 19. http://dx.doi.org/10.1016/j.bone.2013.09.003 ; PubMed PMID: 24056176; PubMed Central PMCID: PMC3856860 , Dec-2013 |
| **Articles in Peer-reviewed Journals** | Prisby RD, Dominguez JM 2nd, Muller-Delp J, Allen MR, Delp MD. "Aging and estrogen status: A possible endothelium-dependent vascular coupling mechanism in bone remodeling." PLoS One. 2012;7(11):e48564. http://dx..doi.org/10.1371/journal.pone.0048564 ; PubMed PMID: 23185266; PubMed Central PMCID: PMC3502426 , Nov-2012 |
| **Articles in Peer-reviewed Journals** | Stabley JN, Moningka NC, Behnke BJ, Delp MD. "Exercise training augments regional bone and marrow blood flow during exercise." Med Sci Sports Exerc. 2014 Nov;46(11):2107-12. http://dx.doi.org/10.1249/MSS.0000000000000342 ; PubMed PMID: 24658222; PubMed Central PMCID: PMC4169763 , Nov-2014 |
| **Articles in Peer-reviewed Journals** | Prisby RD, Behnke BJ, Allen MR, Delp MD. "Effects of skeletal unloading on the vasomotor properties of the rat femur principal nutrient artery." J Appl Physiol (1985). 2015 Apr 15;118(8):980-8. http://dx.doi.org/10.1152/japplphysiol.00576.2014 ; PubMed PMID: 25635000; PubMed Central PMCID: PMC4398884 , Apr-2015 |
| **Articles in Peer-reviewed Journals** | Stabley JN, Prisby RD, Behnke BJ, Delp MD. "Type 2 diabetes alters bone and marrow blood flow and vascular control mechanisms in the ZDF rat." J Endocrinol. 2015 Apr;225(1):47-58. http://dx.doi.org/10.1530/JOE-14-0514 ; PubMed PMID: 25817711; PubMed Central PMCID: PMC4379453 , Apr-2015 |